SEJARAH SEMAKAN					
Terbitan	**Semakan**	**Ditulis oleh**	**Disemak oleh**	**Diluluskan oleh**	**Tarikh kuatkuasa**
1	0	Nurul 'Izzah Hany Ismail	Ooi Suat Hong	Ida Syazrina Ibrahim	15 Nov 2024
1	1	Nurul 'Izzah Hany Ismail	Ooi Suat Hong	Ida Syazrina Ibrahim	1 Julai 2025
2	0	Nurul Nadiah Noh	Tan Lu Yi	Ida Syazrina Ibrahim	10 April 2026

RUJUKAN	
No. Dokumen	**Tajuk**
	ASEAN Harmonised Method ACM 007
PKKK/300/UP/003	Rapid Resolution Liquid Chromatography HP 1200
PKKK/300/UP/011	Shimadzu High Performance Liquid Chromatography (Prominence-i)
PKKK/300/UP/016	High Performance Liquid Chromatography Flexar (Perkin Elmer)
PKKK/300/UP/001	Proses Persampelan (Ujian Penyaringan)
PKKK/300/UP/013	Pengekstrakan Fasa Pepejal Bagi Produk Tradisional, Suplemen Kesihatan dan Kosmetik
PKKK/300/UP/014	Pengekstrakan Fasa Cecair Bagi Produk Tradisional, Suplemen Kesihatan dan Kosmetik

SEJARAH SEMAKAN

Terbitan 1, Semakan 1:

i. Kemaskini 6.6 System suitability test requirement, di mana

a) kritieria *Theoretical plate* telah dikemaskini daripada "(NLT) 1500 kepada "(NLT) 2000".
b) Menambah "%RSD for retention time: Not more than, NMT 2.0%"

ii. Menambah perenggan 6.7 Internal Quality Control.
iii. Mengemaskini perenggan 5.4, penolong pegawai farmasi ditukar daripada U36 / U32 kepada U7 / U6.
iv. Mengemaskini perenggan 3 Definisi
v. Menambah perenggan 6.3.3 Single Standard Stock Solution
vi. Mengemaskini perenggan 6.5.5 Injection Sequence
vii. Mengemaskini perenggan 6.8.1(b) dengan menambah "The UV spectrum is identical to the spectrum of the standard with similarity of more than 0.999 to the spectrum of standard."
viii. Mengemaskini perenggan 6.9g dengan menambah "IQC table"

Terbitan 2, Semakan 0:

i. Meminda No.Dokumen dari PKKK/300/UAT/047 kepada PKKK/300/UP/021
ii. Kemaskini tarikh kuatkuasa dari 1 Julai 2026 kepada 10 April 2026
iii. Kemaskini No. Dokumen rujukan dari Unit Analisis Tradisional kepada Unit Penyaringan:
    a. PKKK/300/UAT/004 kepada PKKK/300/UP/003
    b. PKKK/300/UAT/012 kepada PKKK/300/UP/011
    c. PKKK/300/UAT/017 kepada PKKK/300/UP/016
    d. PKKK/300/UAT/001 kepada PKKK/300/UP/001
    e. PKKK/300/UAT/014 kepada PKKK/300/UP/013
    f. PKKK/300/UAT/015 kepada PKKK/300/UP/014

1. TUJUAN

Untuk memastikan ujian identifikasi steroid menggunakan alat *High Performance Liquid Chromatography* (HPLC) dilaksanakan dengan cekap dan berkesan

2. SKOP

Prosedur ini hendaklah digunakan untuk menjalankan ujian identifikasi steroid dalam produk tradisional menggunakan HPLC.

3. DEFINISI

3.1	HPLC	- High Performance Liquid Chromatography
3.2	RS	- Reference Standard
3.3	OOS	- Out of Specification
3.4	UV	- Ultraviolet
3.5	SST	- System Suitability Test
3.6	IQC	- Internal Quality Control
3.7	RSD	- Relative Standard Deviation
3.8	PDA	- Photodiode Array
3.9	RRLC	- Rapid Resolution Liquid Chromatography

4. CARTA ALIRAN

Tiada carta aliran untuk prosedur ini.

5. TANGGUNGJAWAB

5.1 Ketua Seksyen

- Memastikan prosedur ini dipatuhi semasa pengujian dijalankan
- Menyemak dan mengesahkan semua laporan ujian tidak lulus yang telah disemak oleh Pegawai Farmasi termasuk laporan OOS

5.2 Ketua Unit / Pegawai Farmasi Bertanggungjawab

- Memastikan prosedur ini dipatuhi semasa pengujian dijalankan

- Menyemak dan mengesahkan semua laporan ujian yang telah disemak oleh Pegawai Farmasi termasuk laporan OOS

5.3 Pegawai Farmasi

- Mengagihkan sampel kepada Pegawai Farmasi dan Penolong Pegawai Farmasi
- Menyelia ujian yang dijalankan oleh Pegawai Farmasi dan Penolong Pegawai Farmasi
- Menyemak laporan ujian
- Menjalankan siasatan OOS dan menyediakan laporan OOS bagi sampel gagal ujian
- Menjalankan aktiviti pengujian sampel mengikut tatacara yang telah ditetapkan

5.4 Penolong Pegawai Farmasi U7 / U6

- Merekodkan bilangan ujian selepas selesai laporan ujian disiapkan untuk tujuan laporan bulanan
- Menghantar laporan ujian ke Unit Perkhidmatan Analisis untuk simpanan

5.5 Penolong Pegawai Farmasi

- Menjalankan aktiviti pengujian sampel mengikut tatacara yang telah ditetapkan

6. PROSEDUR

6.1 Materials and Equipment

i. Instrument:
   - High Performance Liquid Chromatography with photodiode array (PDA) detector

ii. Analytical & microbalance

iii. Column: Kinetex Phenyl-Hexyl 2.6µm, 150 x 4.6 mm or equivalent

iv. Reference standards: Primary or secondary RS or equivalent
   - Prednisolone

- Prednisone
- Betamethasone
- Dexamethasone
- Triamcinolone acetonide
- Hydrocortisone acetate
- Cortisone acetate
- Betamethasone-17-valerate

v. Chemicals:

- Ethanol
- Acetonitrile
- Deionized water
- Ammonia
- Methanol

vi. Glassware and consumables:

- Beakers (50 mL)
- Solvent filter system
- Nylon solvent filter 0.45 µm
- Solvent bottle (1 L)
- Nylon syringe filter 0.45 µm
- Volumetric Flasks (Grade A)
- Disposable syringe without needle (5 mL)
- 2 mL glass screw cap vial

6.2 Preparation of Mobile Phase

i. Measure 1 L of water and transfer in a solvent bottle.
ii. Repeat step i. for acetonitrile.
iii. Filter water and acetonitrile using nylon solvent filter 0.45 µm and solvent filter system
iv. Degassed filtered mobile phase at least for 20 minutes to ensure no bubble present.

6.3 Preparation of Reference Standard

6.3.1 Individual standard stock solution

Weigh 10 mg Prednisolone, Prednisone, Betamethasone, Dexamethasone, Triamcinolone acetonide, Hydrocortisone acetate, Cortisone acetate and Betamethasone-17-Valerate standard in 10 mL volumetric flask separately. Make up to volume with ethanol. The final concentration of individual solution is 1 mg/mL.

6.3.2 Mix working standard solution

Prepare a mixed working standard solution of 0.05 mg/mL in concentration by transferring 1 mL of each standard stock solution into one volumetric flask (20 mL) and make up to volume with ethanol. Mix well.

6.3.3 Single Standard Stock Solution

6.3.3.1 Prepare a single standard stock solution following the same procedure outlined in 6.3.1 to 6.3.2, but using a single standard instead of multiple standards.

6.3.3.2 This procedure is applicable for screening selected compounds only

6.4 Preparation of sample

6.4.1 Sample handling (sampling)

Sampling processes differ between different type of dosage form. Please refer to the document PKKK/300/UP/001 (Proses Persampelan).

6.4.2 Sample preparation for analysis

- Sample preparation differ between dosage forms. Please refer to document PKKK/300/UP/013 (Pengekstrakan Fasa Pepejal Bagi Produk Tradisional, Suplemen Kesihatan dan Kosmetik) and PKKK/300/UP/014 (Pengekstrakan Fasa Cecair Bagi Produk Tradisional, Suplemen Kesihatan dan Kosmetik).
- Sample pH is adjusted to pH7 before the extraction process is performed.

6.5 Experimental Method

6.5.1 The HPLC system used for identification of steroids are given in Table 1 below.

Column	Kinetex Phenyl-Hexyl 100A or equivalent
Dimensions & Particle size	2.6µm, 150 x 4.6 mm
Flow rate	0.7 mL/min
Column temperature	40°C
Wavelength detector (PDA)	240 nm
Injection volume	10 µL
Run time	35 minutes
Mobile Phase Method	Gradient program

Table 1: HPLC system used for identification of steroids.

6.5.2 Mobile Phase Method

Solvent gradient program for mobile phase is use in identification of steroids shown as below:

Time (min)	Water (%)	Acetonitrile (%)
0.01 – 1.99	80	20
2.00 - 13.99	75	25
14.00 - 17.99	70	30
18.00 – 19.99	60	40
20.00 – 21.99	50	50
22.00 - 29.99	30	70
30.00 – 35.00	80	20

Table 2: Solvent gradient program for identification of steroids.

Refer to document PKKK/300/UP/003 (RRLC 1200 Series), PKKK/300/UP/011 (HPLC Prominence-i) and PKKK/300/UP/016 (HPLC Flexar) on how to set-up method and HPLC system.

6.5.2 Standard Injection and System Suitability

Mix working standard solution must be injected at the beginning of analysis (six injections). Repeat the injection or prepare a new mix standard solution if the chromatogram of reference standard is not well separated (except betamethasone & dexamethasone) or peak is not sharp.

6.5.3 Sample Injection

Inject each of the sample in duplicate. Compare the retention time and UV spectrum of the peak in the chromatogram of the sample with the reference standard. The sample contains steroid compounds if the retention time and UV spectrum of the sample correspond to the reference standard.

6.5.4 Retention time and UV spectrum observation

To observe and check retention time and UV spectrum of the sample and RS, refer to document PKKK/300/UP/003 (RRLC 1200 Series), PKKK/300/UP/011 (HPLC Prominence-i) and PKKK/300/UP/016 (HPLC Flexar).

6.5.5 Injection Sequence

i. First, inject a blank or diluent, then inject a mix working standard solution six times (SST) before injecting the sample.
ii. Inject the sample in duplicate.
iii. If more than five samples are injected in a single analysis, the mix working standard solution must be injected once every five samples.
iv. A blank or diluent must be injected before the next sample injection. Refer to Table 3 for an example of the sequence table:

No.	Detail	No. of injection
1	Blank / Diluent	1
2	Mix working standard solution (for SST)	6
3	Sample number 1	2
4	Blank / Diluent	1
5	Sample number 2	2
6	Blank / Diluent	1
7	Sample number 3	2
8	Blank / Diluent	1
9	Sample number 4	2
10	Blank / Diluent	1
11	Sample number 5	2
12	Blank / Diluent	1

13	Mix working standard solution	1
14	Blank / Diluent	1
15	Sample number 6	2
16	Blank / Diluent	1
17	Mix working standard solution (for SST)	1

Table 3: Sequence injection for identification of steroid in HPLC

v. Mix working standard solution must be injected at the end of the analysis.

6.6 System suitability test requirement

- System suitability tests need to be performed for each batch run to ensure the entire system is working properly. Perform at least six replicate injections of mix working standard solution (0.05 mg/mL).
- Acceptance criteria:
  %RSD for retention time: Not more than, NMT 2.0%
  Tailing factor not more than (NMT) 2.0.
  Theoretical plate not less than (NLT) 2000.

6.7 Internal Quality Control (IQC)

6.7.1 Inject 10 µL of mix standard solution after every 5 samples & at the end of analysis.

6.7.2 Acceptance criteria for Internal Quality Control Check (IQC):
The percentage difference of retention time for a mixed standard solution should not be more than 3 % for every batch of 5 samples.

6.8 Acceptance criteria for test sample

6.8.1 Compare the retention time and UV spectrum obtained from the sample chromatogram with that of the working standard chromatogram. The identity of each steroid compound is positive if:

a) Retention time of the compound in the sample chromatogram corresponds to the retention time of the same compound in the standard chromatogram.

b) The UV spectrum is identical to the spectrum of the standard with similarity of more than 0.999 to the spectrum of standard.

6.8.2 If the sample was found to be positive with one of the compounds, new sample preparation should be conducted by a second analyst on a different date from the first analyst.

6.8.3 The second analyst will prepare a new sample following 6.3.2 in the guidelines outlined in PKKK/300/UP/014 (Pengekstrakan Fasa Cecair Bagi Produk Tradisional, Suplemen Kesihatan dan Kosmetik). The pH of the sample will need to be adjusted to pH 7.

6.8.4 The sample will then be analyzed using GCMS (PKKK/300/UP/004, Gas Chromatography Mass Spectroscopy Agilent 7890A/5975C or PKKK/300/UP/017, Gas Chromatography Mass Spectroscopy Agilent 8890/5977B).

6.9 Analysis report

Each sample will have one analysis report. Report must consist of detail as below:

a. Sample details (Sample name, sample number and batch number)
b. Analysis date
c. Equipment parameters including equipment name, mobile phase program and HPLC system.
d. List and preparation procedures of reference solution use in analysis.
e. Sample preparation procedure and printed sample weight (for solid sample only)
f. Interpretation of result (Detected, Not Detected or suspected of specific steroid compound)
g. Evidence of analysis must be attached together in analysis report:
    - Chromatogram of 1st and 2nd injection of sample (with minimum information of compound name, peak area and retention time).
    - Chromatogram of 1st and 2nd injection of standard (with minimum information of compound name, peak area and retention time). If the sample is sequenced at 6 or more, a chromatogram of standard before sample must be presented.
    - Overlay spectrum of samples that have similar retention time with reference standard.
    - IQC table

7. REKOD KUALITI

Tiada rekod kualiti bagi prosedur ini.

LAMPIRAN (1/4)

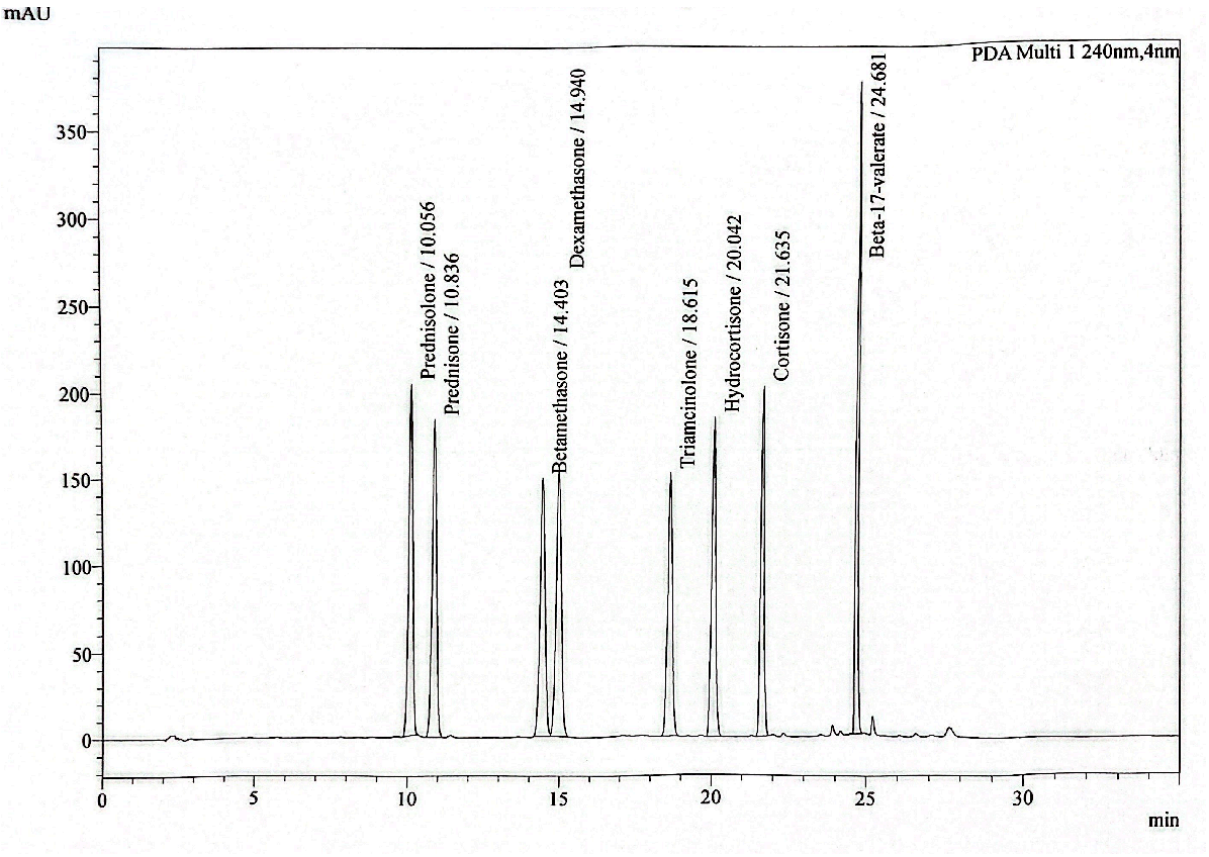
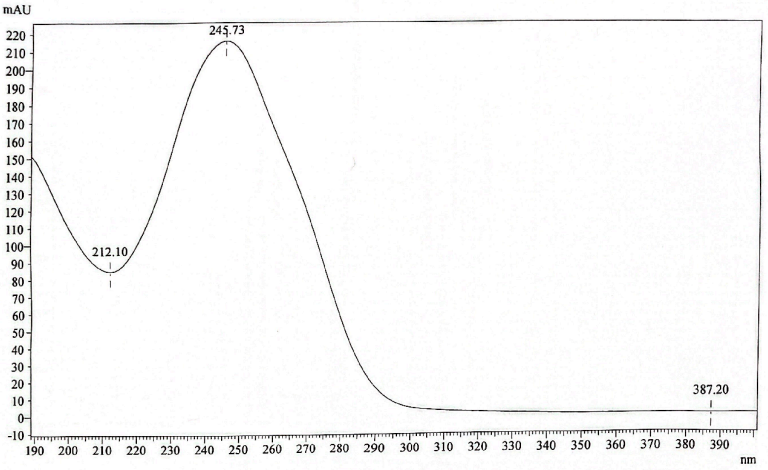


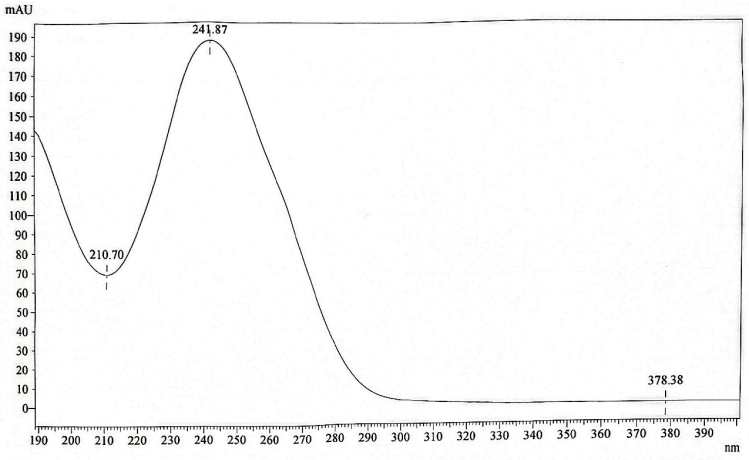
Figure 1: Chromatogram of mix working standard solution at 240nm.

LAMPIRAN (2/4)

a)



b)



c)

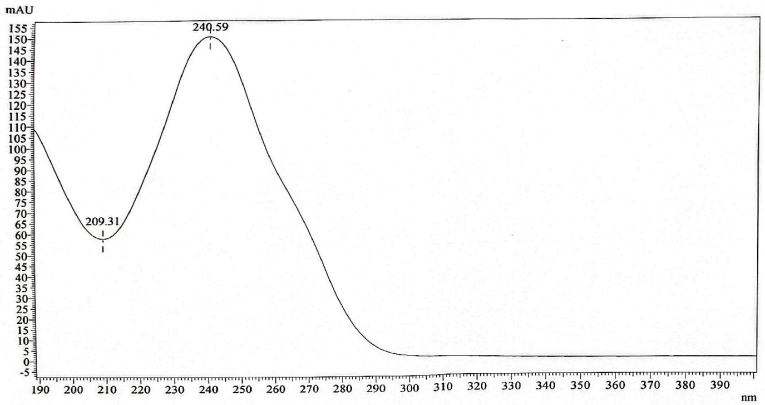
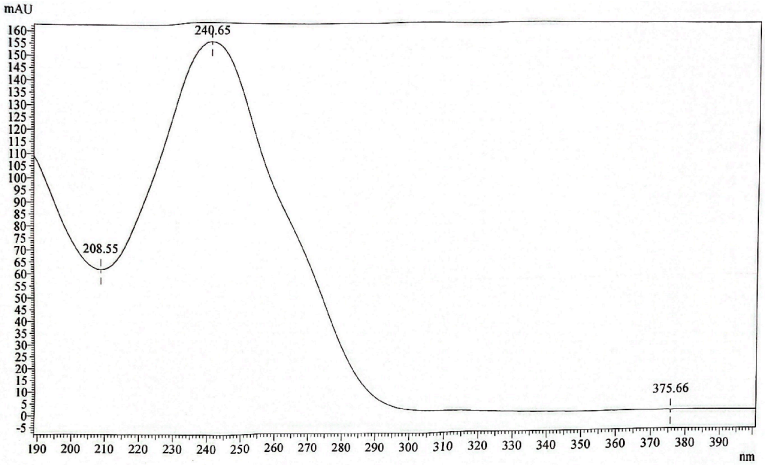


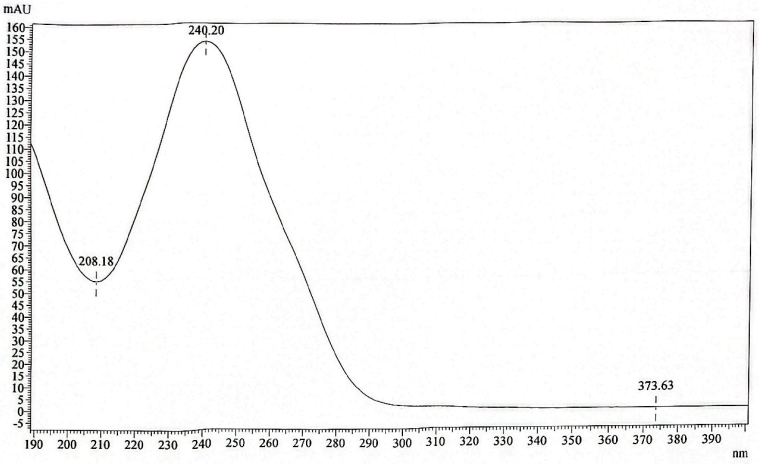
Figure 2: Spectra of a) Prednisone, b) Prednisolone and c) Betamethasone in scan range 100-400nm.

LAMPIRAN (3/4)

d)



e)



f)

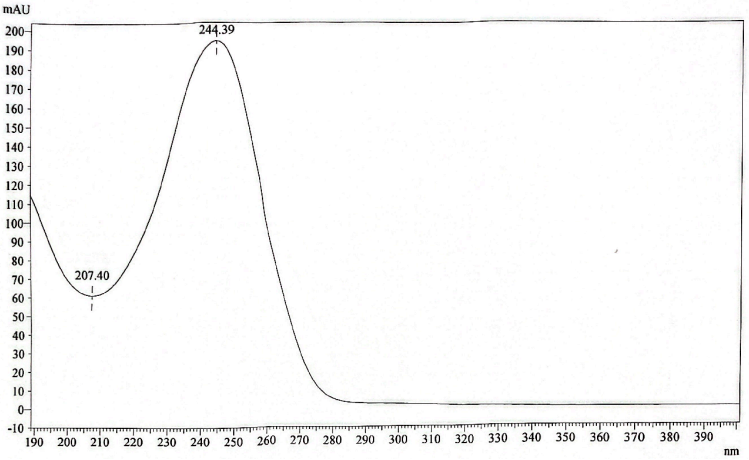
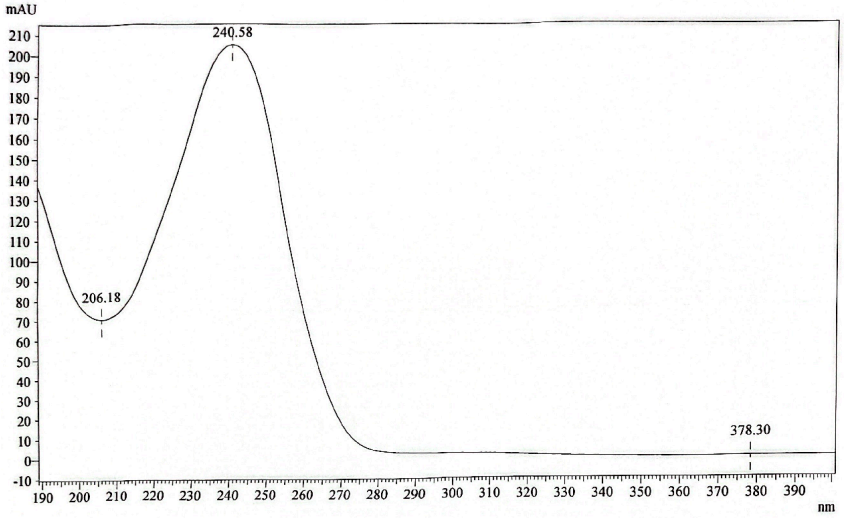


Figure 3: Spectra of d) Dexamethasone, e) Triamcinolone acetonide and f) Hydrocortisone acetate in scan range 100-400nm.

LAMPIRAN (4/4)

g)



h)

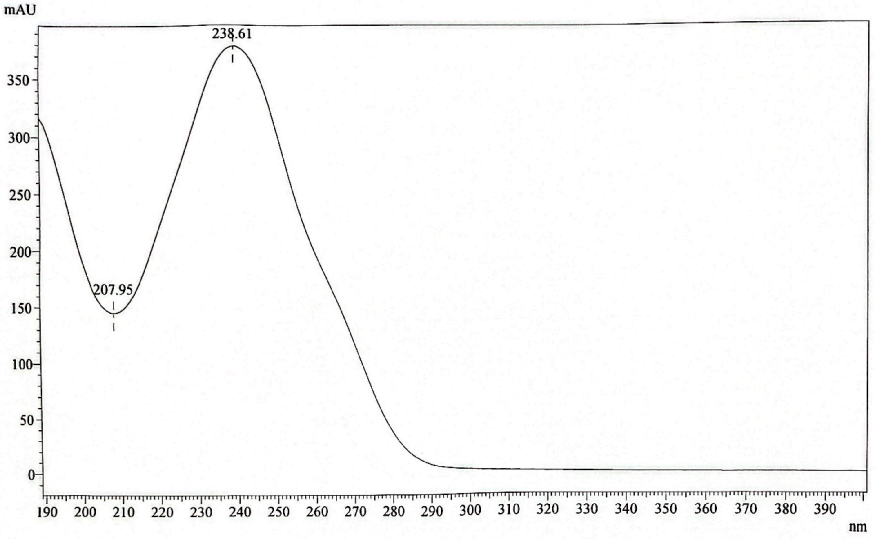


Figure 4: Spectra of g) Cortisone acetate and h) Beta-17-valerate in scan range 100-400nm.